5.4.4 Limits of monotone functions

The situation is better for monotone functions, just as it was for monotone sequences:

Theorem 5.6: A monotone (increasing or decreasing) function $f : \mathbb{R} \to \mathbb{R}$ cannot have a discontinuity of the second type. That is, a monotone function could only have removable discontinuities, jump discontinuities, or have asymptotes (vertical or horizontal).

Proof. We prove the theorem for a monotone increasing function. The same ideas will carry over for a monotone decreasing function. We split the proof into two claims:

(1) Claim: for any $x_0 \in \{-\infty\} \cup \mathbb{R}$,

$$\lim_{x \to x_0^+} f(x) = \inf_{x > x_0} f(x).$$

Let $L_+ = \inf_{x>x_0} f(x)$ and suppose that $L_+ \in \mathbb{R}$. By the definition of the infimum, for any $\varepsilon > 0$, there exists $x_1 > x_0$ such that $f(x_1) < L_+ + \varepsilon$. In the case $x_0 \in \mathbb{R}$ define $\delta = \delta(\varepsilon) = x_1 - x_0 > 0$, and in the case $x_0 = -\infty$ define $A = A(\varepsilon) = |x_1|$. Since f is monotone increasing, for all $x \in (x_0, x_1)$, we have $L_+ \leq f(x) \leq f(x_1) < L_+ + \varepsilon$. Thus, in the case $x_0 \in \mathbb{R}$, we have $|f(x) - L_+| < \varepsilon$ whenever $0 < x - x_0 < \delta$, and in the case $x_0 = -\infty$, we have $|f(x) - L_+| < \varepsilon$ whenever $x < -A$. In either case, this proves that the right-hand limit exists and equals L_+. A similar idea proves the claim for $L_+ = -\infty$.

(2) Claim: for any $x_0 \in \mathbb{R} \cup \{+\infty\}$,

$$\lim_{x \to x_0^-} f(x) = \sup_{x < x_0} f(x).$$

Let $L_- = \sup_{x<x_0} f(x)$ and suppose that $L_- \in \mathbb{R}$. By the definition of the supremum, for any $\varepsilon > 0$, there exists $x_1 < x_0$ such that $f(x_1) > L_- - \varepsilon$. In the case $x_0 \in \mathbb{R}$ define $\delta = \delta(\varepsilon) = x_0 - x_1 > 0$, and in the case $x_0 = +\infty$ define $A = A(\varepsilon) = |x_1|$. Since f is increasing, for all $x \in (x_1, x_0)$, we have $L_- - \varepsilon < f(x_1) \leq f(x) \leq L_-$. Thus, in the case $x_0 \in \mathbb{R}$, we have $|f(x) - L_-| < \varepsilon$ whenever $0 < x_0 - x < \delta$, and in the case $x_0 = +\infty$, we have $|f(x) - L_-| < \varepsilon$ whenever $x > A$. In either case, this proves that the left-hand limit exists and equals L_-. A similar idea proves the claim for $L_- = +\infty$.

Hence, at any point $x_0 \in \mathbb{R}$, both one-sided limits exist (though they may be infinite). The only possibilities are:

- **Removable discontinuity**: when $L_- = \lim_{x \to x_0^-} f(x) = \lim_{x \to x_0^+} f(x) = L_+$.
- **Jump discontinuity**: when $L_- = \lim_{x \to x_0^-} f(x) < \lim_{x \to x_0^+} f(x) = L_+$
- **Vertical asymptote**: when one of the one-sided limits is infinite (then the other one will not exist because of monotonicity): $L_+ = -\infty$ or $L_- = +\infty$.

(*Can you think why it is impossible to have* $L_+ < L_-$*?*) Therefore, a discontinuity of the second type cannot occur. □

Corollary 5.7: Let $f : \mathbb{R} \to \mathbb{R}$ be monotone increasing. Then for any $x_0 \in \mathbb{R}$, if f is defined in a deleted neighborhood of x_0,

$$\lim_{x \to x_0^-} f(x) \leq \lim_{x \to x_0^+} f(x)$$

If f is defined at x_0, then

$$\lim_{x \to x_0^-} f(x) \leq f(x_0) \leq \lim_{x \to x_0^+} f(x).$$

An analogous statement holds for a monotone decreasing function.

Proof. This is an immediate consequence of Theorem 5.6. □

Chapter 6

Properties and computation of limits

6.1 Uniqueness of the limit and local sign of a function

Uniqueness

We always write *the* limit, not *a* limit. Implicitly, we say that it is *unique*. This is true, however it requires proof. Here is the formal statement (an analogous statement could be made for sequences):

Theorem 6.1 (Uniqueness of limits)**:** Let $f : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$. Suppose that $\lim_{x\to x_0} f(x) = \ell$, where ℓ could be finite or infinite. Then there can be no limit other than ℓ as $x \to x_0$.

Proof. Exercise. *Hint: by contradiction.* □

Local sign

It is intuitively clear that if a function has a positive limit (or $+\infty$), then as we approach this limit the values of the function must also be positive. Analogously, if a limit is negative (or $-\infty$), then the values nearby should be negative. This is stated as follows:

Theorem 6.2 (Local sign)**:** Let $f : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \mathbb{R}$.

- If $\lim_{x\to x_0} f(x) > 0$ or $\lim_{x\to x_0} f(x) = +\infty$
  then $f > 0$ on a deleted neighborhood of x_0.
- If $\lim_{x\to+\infty} f(x) > 0$ or $\lim_{x\to+\infty} f(x) = +\infty$
  then there exists $M > 0$ s.t. $f > 0$ on $\{x > M\}$.
- If $\lim_{x\to-\infty} f(x) > 0$ or $\lim_{x\to-\infty} f(x) = +\infty$
  then there exists $M < 0$ s.t. $f > 0$ on $\{x < M\}$.

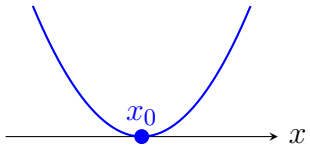
Analogous statements hold if these limits are negative.

Proof. We prove the first claim: $\lim_{x\to x_0} f(x) > 0 \quad \Rightarrow \quad f > 0$ on a deleted neighborhood of x_0. Let $\ell = \lim_{x\to x_0} f(x) > 0$. Let $\varepsilon = \frac{\ell}{2} > 0$. By the definition of the limit, there exists $\delta = \delta(\varepsilon) > 0$ such that for $0 < |x - x_0| < \delta$

$$f(x) \in (\ell - \varepsilon, \ell + \varepsilon) = \left(\ell - \frac{\ell}{2}, \ell + \frac{\ell}{2}\right) = \left(\frac{\ell}{2}, \frac{3\ell}{2}\right) \subset (0, +\infty).$$

Hence $f > 0$ on this neighborhood of x_0 (potentially excluding x_0 itself), which completes the proof. The other claims in the theorem are proved in a similar way. □

The converse of this theorem is *almost* true. As the figure below shows, we can have situations where for all $x \in I_\delta^*(x_0)$ (for some $\delta > 0$ small), $f(x) > 0$, and yet $f(x_0) = 0$.



Hence we can prove the following statement, which is not quite the converse of the previous theorem:

Theorem 6.3: Let $f : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \mathbb{R}$. Assume that $\lim_{x\to x_0} f(x)$ exists.

- If $f \geq 0$ on a deleted neighborhood of x_0
  then $\lim_{x\to x_0} f(x) \geq 0$ or $\lim_{x\to x_0} f(x) = +\infty$.
- If there exists $M > 0$ s.t. $f \geq 0$ on $\{x > M\}$
  then $\lim_{x\to+\infty} f(x) \geq 0$ or $\lim_{x\to+\infty} f(x) = +\infty$
- If there exists $M < 0$ s.t. $f \geq 0$ on $\{x < M\}$
  then $\lim_{x\to-\infty} f(x) \geq 0$ or $\lim_{x\to-\infty} f(x) = +\infty$.

Analogous statements hold if these limits are negative.

Proof. We prove the first claim (the others follow a similar strategy). By contradiction, assume that $f \geq 0$ on a deleted neighborhood of x_0 and that $\lim_{x\to x_0} f(x) < 0$ or $\lim_{x\to x_0} f(x) = -\infty$. We immediately obtain a contradiction to Theorem 6.2. □

Theorem 6.4 (Local boundedness)**:** Let $f : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \mathbb{R}$.

- If $\lim_{x\to x_0} f(x)$ exists and is finite, then f is bounded on a deleted neighborhood of x_0: there exist $\delta > 0$ and $A > 0$ such that for all $0 < |x - x_0| < \delta$, $|f(x)| < A$.
- If $\lim_{x\to+\infty} f(x)$ exists and is finite, then f is bounded for all large x: there exist $A > 0$ and $M > 0$ such that for all $x > M$, $|f(x)| < A$.
- If $\lim_{x\to-\infty} f(x)$ exists and is finite, then f is bounded for all large negative x: there exist $M < 0$ and $A > 0$ such that for all $x < M$, $|f(x)| < A$.

Proof. We prove the first claim. Denote $\ell = \lim_{x\to x_0} f(x) \in \mathbb{R}$. By definition of the limit, for any $\varepsilon > 0$ there exists $\delta > 0$ such that for any $0 < |x - x_0| < \delta$, we have $f(x) \in (\ell-\varepsilon, \ell+\varepsilon)$. Choosing $A = |\ell|+\varepsilon$ will do the job: $|f(x)| < A$ for all $x \in I_\delta^*(x_0)$. □